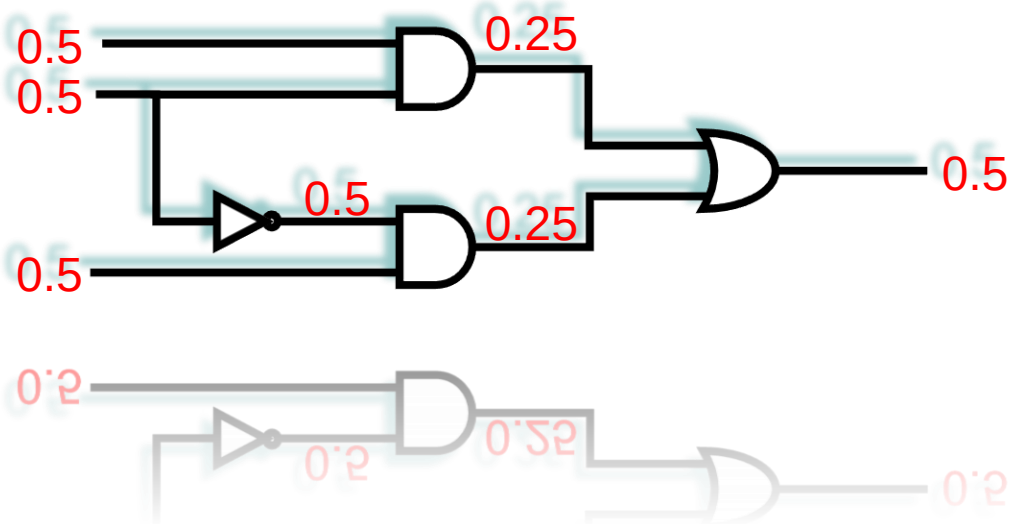


Testability Measure

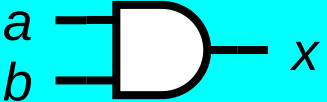
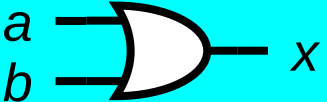
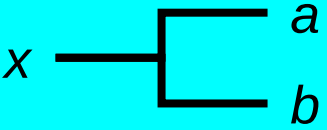
- Introduction
- SCOAP (1979)
- **COP (1984)**
- High-level testability measures
- Conclusion



COP

- **Signal probability of x** = probability of x being logic 1
 - ♦ Actual signal probability requires exhaustive simulation
 - ♦ Hard to obtain in practice
- **COP** = Controllability/Observability Program [Brglez 84]
 - ♦ Fast algorithm to estimate signal probability
 - ♦ C_x = estimated $\text{prob}(x = 1)$
 - ♦ $1 - C_x$ = estimated $\text{prob}(x = 0)$
 - ♦ O_x = estimated probability of *fault effect* in x being observed at PO
- C_x and O_x are numbers between 0 and 1
 - ♦ *Larger number* means *easier* to control or observe
- **Assumptions**
 - ♦ 1. Ignore fanout reconvergence for fast run time
 - ♦ 2. PI are independent random numbers: $\frac{1}{2}$ zero and $\frac{1}{2}$ one

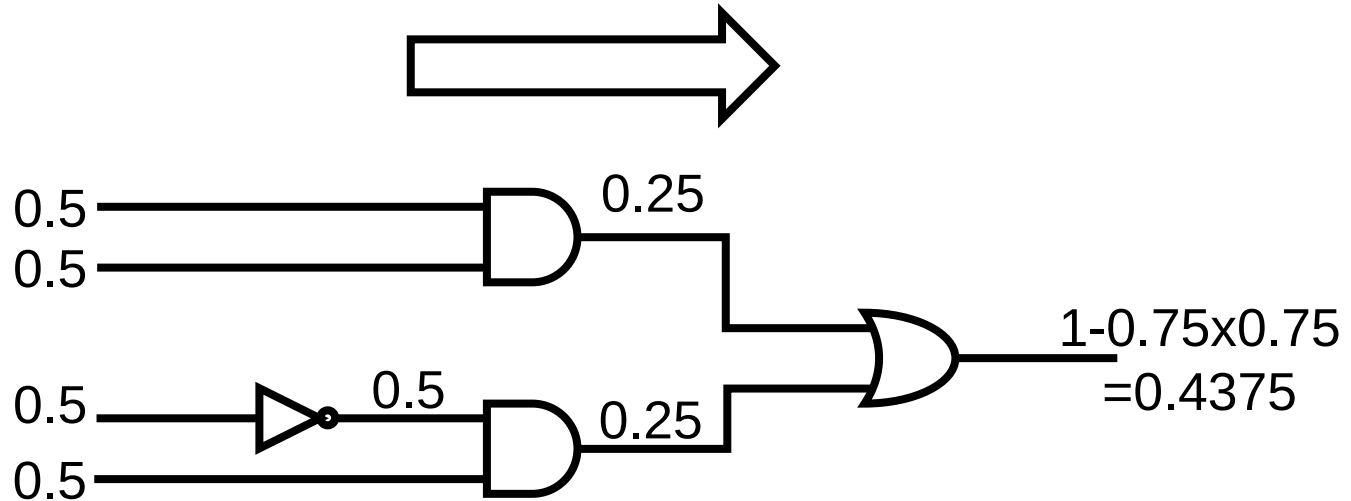
COP

	C_x	O_a
$x = PI$	0.5	
$x = PO$		0.5
	$C_x = C_a \times C_b$	$O_a = O_x \times C_b$
	$C_x = 1 - (1 - C_a) \times (1 - C_b)$	$O_a = O_x \times (1 - C_b)$
	$C_x = C_a = C_b$	$O_x = 1 - (1 - O_a) \times (1 - O_b)$

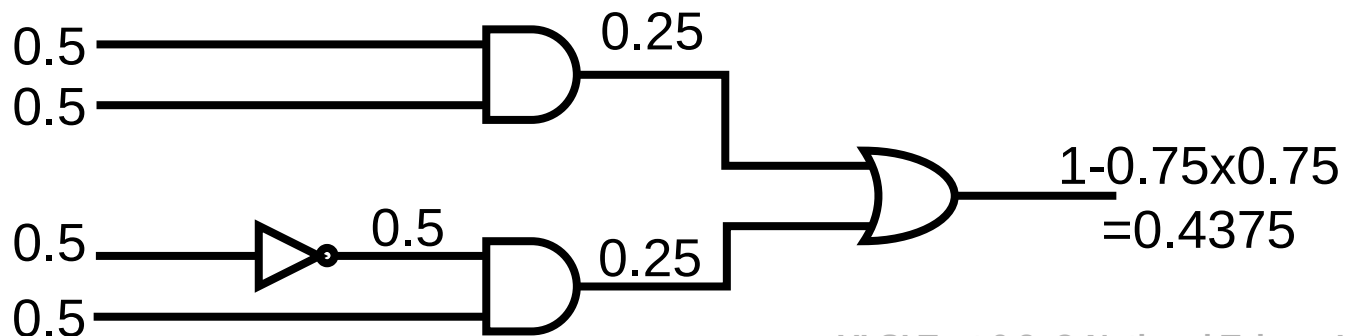
Example – Controllability

- Calculate from PI to PO
- **Fanout-free circuit**, COP = actual signal probability

COP C_x



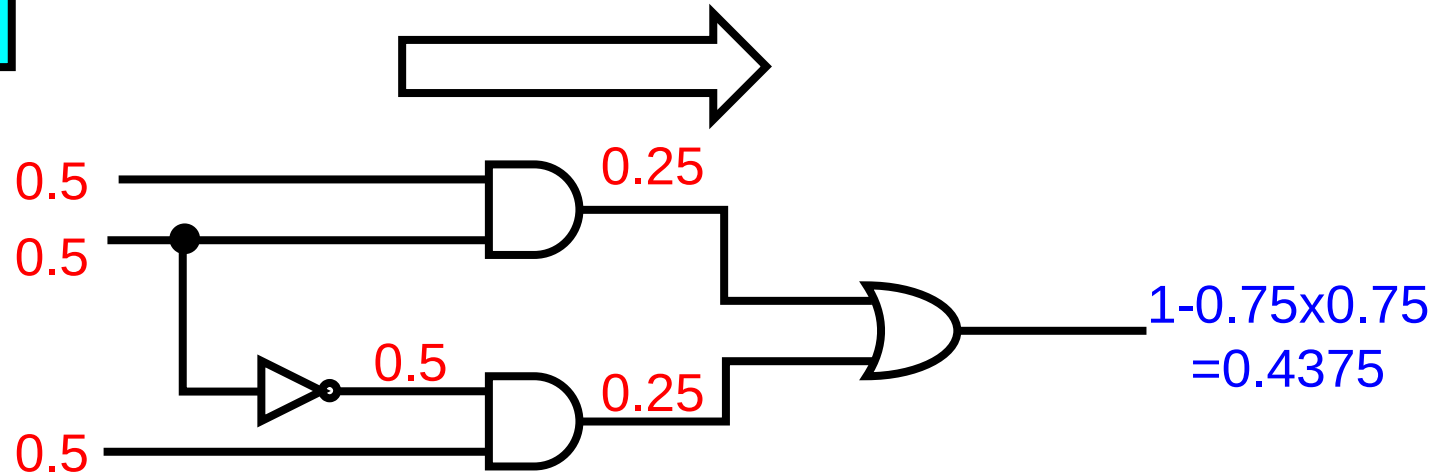
Actual signal probability



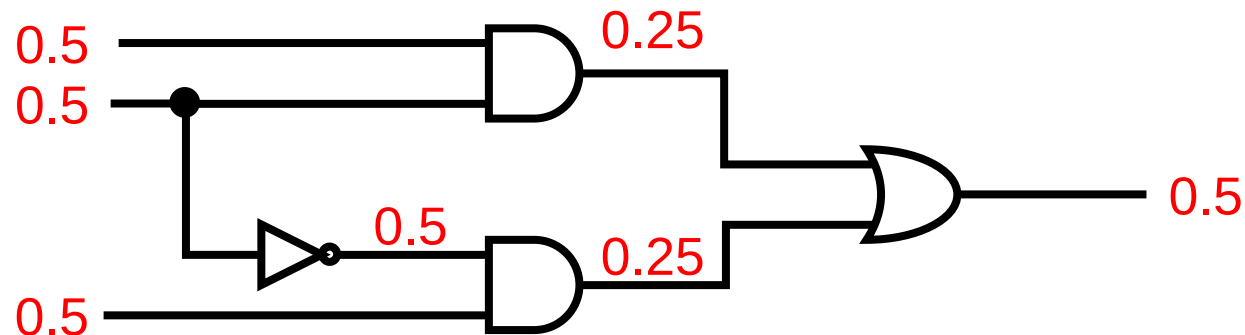
Example (2) – Controllability

- When fanouts reconverge, COP $\neq$ actual signal probability

COP C_x



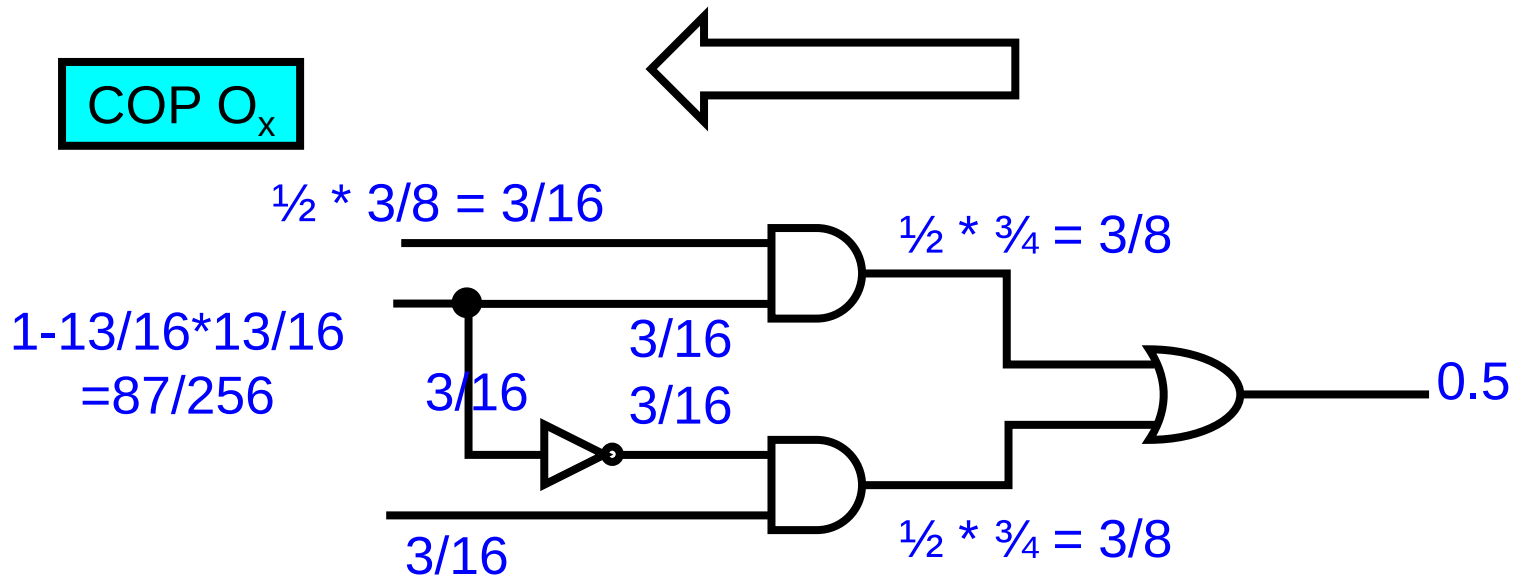
Actual signal probability



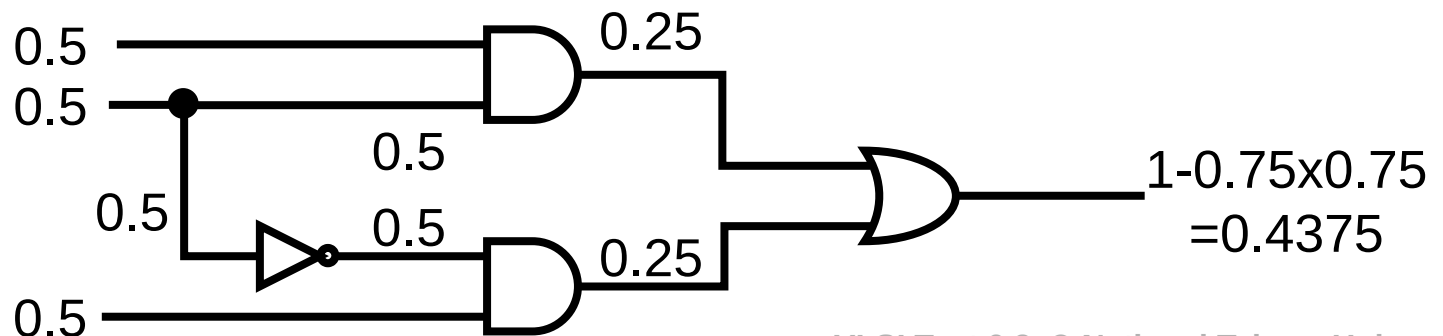
Example (2) – Observability

- Calculate from PO to PI

COP O_x



COP C_x

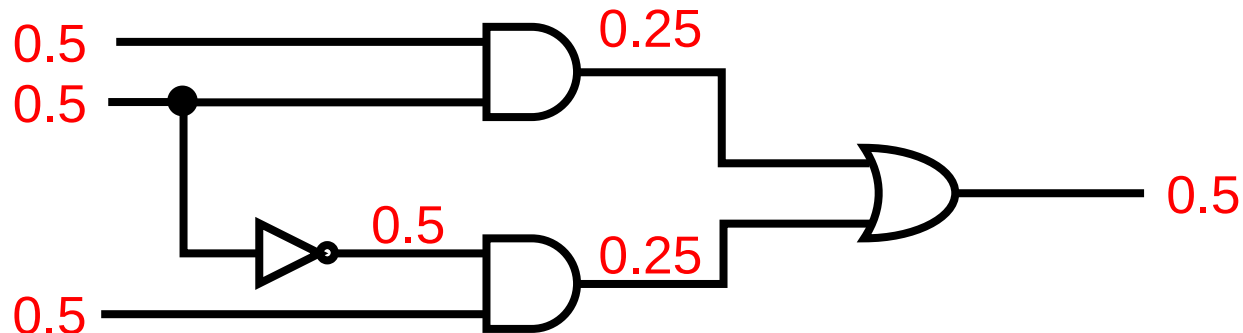


Quiz

Q: verify actual signal probability by exhaustive test patterns

input	output
000	
001	
010	
011	
100	
101	
110	
111	

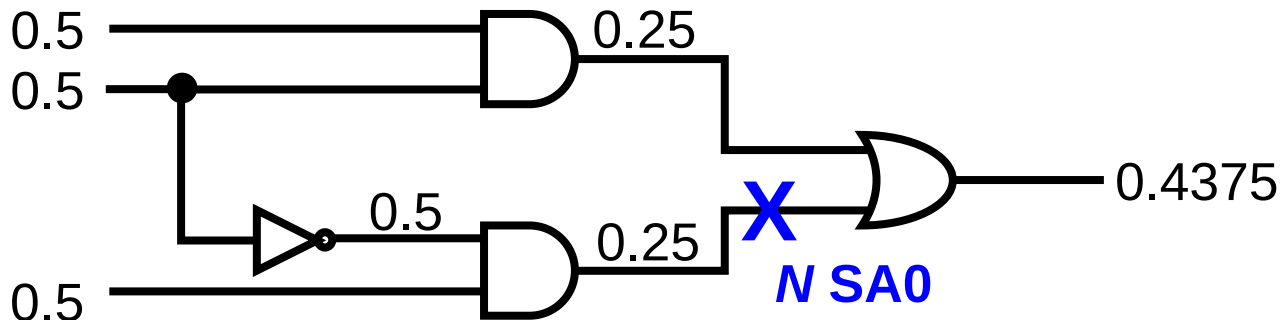
Actual signal probability



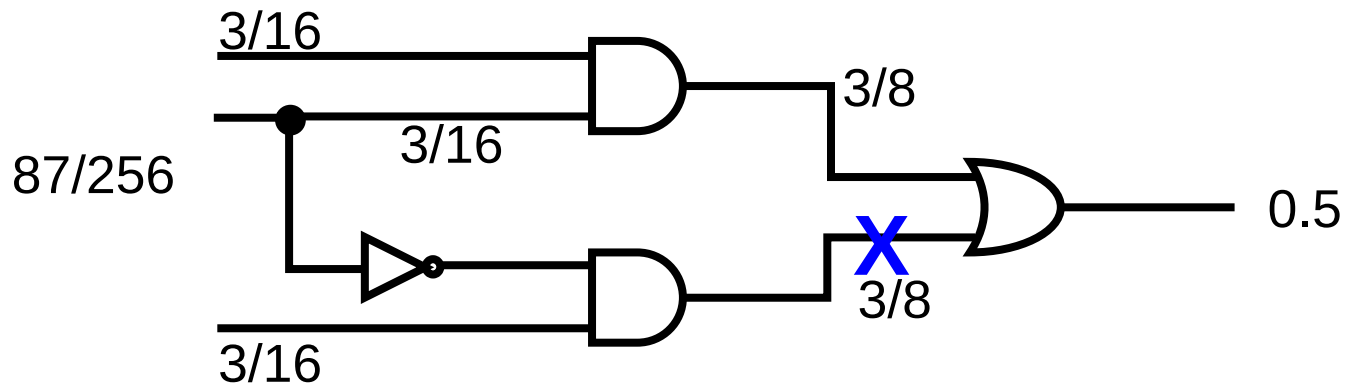
Detection Probability, DP

- DP_f = Probability of detecting a fault f
 - ♦ $DP_{N\ SA0} = C_N \times O_N$
 - ♦ $DP_{N\ SA1} = (1 - C_N) \times O_N$
- Larger DP_f means *easier* to detect fault f
- Example: $DP_{N\ SA0} = 1/4 \times 3/8 = 3/32$

C_x



O_x



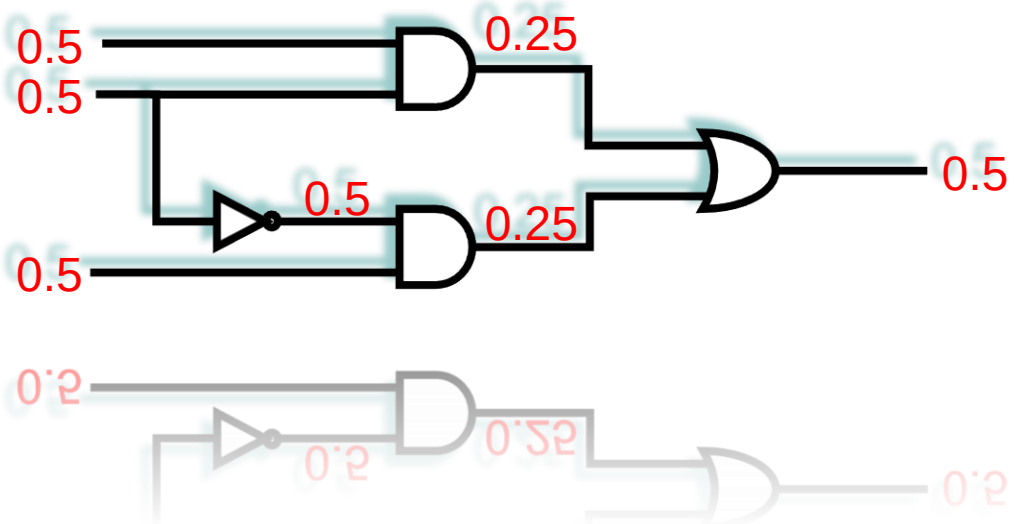
Random Pattern Resistant Faults

- **RPRF** = Faults that are difficult to be tested by random patterns
 - ◆ *Low detectability*
 - ◆ aka. **Hard-to-detect faults, difficult faults**
- Example:
 - ◆ stuck-at-0 fault at an n -input AND gate output
 - ◆ Need test pattern $(1,1,1,\dots,1)$
 - ◆ Assume equal signal probability of 0.5 at each input
 - * $C_x = 0.5^n$
- Test generation for RPRF is difficult
 - ◆ Solutions:
 - * 1. Insert test points (See DFT lecture)
 - * 2. Weighted random patterns (see BIST lecture)

Summary

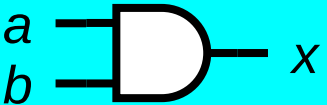
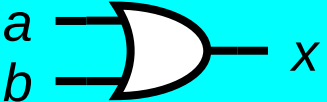
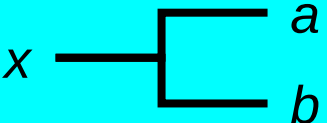
- COP

- ◆ C_x = estimated prob($x = 1$)
- ◆ $1 - C_x$ = estimate prob($x = 0$)
- ◆ O_x = estimated probability of *fault effect* in x being observed
- ◆ **COP $\neq$ actual signal probability** because fanout reconvergence



FFT

- Q: Why observability at PO is 0.5, not 1?

	C_x	O_a
$x = PI$	0.5	
$x = PO$		0.5
	$C_x = C_a \times C_b$	$O_a = O_x \times C_b$
	$C_x = 1 - (1 - C_a) \times (1 - C_b)$	$O_a = O_x \times (1 - C_b)$
	$C_x = C_a = C_b$	$O_x = 1 - (1 - O_a) \times (1 - O_b)$